MILITARY SPECIFICATION

TESTER, HOSE ASSEMBLIES; HYDRAULIC, AIRCRAFT AND MISSILE

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers one kind of air powered, hydraulic tester intended to proof pressure test medium pressure, flared hydraulic hose assemblies (MS28741-3 to -32) and high pressure, flareless hydraulic hose assemblies (MS28762-4 to -16).

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications and standards. Unless otherwise specified (see 6.2) the following specifications and standards of the issue listed in that issue of the Department of Defense Index of Specifications and Standards (DoDISS) specified in the solicitation, form a part of this specification to the extent specified herein.

SPECIFICATIONS

FEDERAL

GG-G-76	Gage, Pressure And Vacuum, Dial Indicating (For Air, Steam, Oil, Water, Ammonia And Chloro-fluoro hydrocarbon Gases And Compressed Gases)
PPP-B-601	Box, Wood, Cleated, Plywood
PPP-B-636	Boxes, Shipping, Fiberboard

MILITARY

MIL-P-116	Preservation, Method of
MIL-P-5510	Packing, Preformed, Straight Thread, Tube Fitting, Boss, Type 1 Hydraulic (Minus 65 Deg. To 160 Deg F)
MIL-G-5514	Gland Design, Packings, Hydraulic, General Requirements For
MIL-P-5516	Packing, Preformed, Petroleum Hydraulic Fluid Resistant, 160 Degree F

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be used in improving this document should be addressed to: the Engineering Division, San Antonio Air Logistics Center/MMEDO, Kelly Air Force Base, Texas 78241 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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MIL-H-5606	Hydraulic Fluid, Petroleum Base, Aircraft, Missile And Ordnance
MIL-P-7105	Pipe Threads, Taper, Aeronautical National Form, Symbol Ampt, General Requirements For
MIL-S-7742	Screw Threads; Standard, Optimum Selected Series, General Specification For
MIL-S-8512	Support Equipment, Acronautical, Special, General Specification For The Design Of

STANDARDS

MILITARY

MIL-STD-129	Marking For Shipment And Storage
MIL-STD-810	Environmental Test Methods and Engineering Guidelines
MIL-STD-831	Test Reports, Preparation Of
MIL-STD-1186	Cushioning, Anchoring, Bracing, Blocking, And Waterproofing With Appropriate Test Methods
MS28741	Hose Assembly, Detachable End Fittings, Medium Pressure
MS28762	Hose Assembly, Hydraulic And Pneumatic, High Pressure, (3000 PSI) Flareless Tube

AIR FORCE - NAVY AERONAUTICAL

AN6236	Filter Element And Hydraulic Replaceable Micronic Reservoir Type

AIR FORCE - NAVY AERONAUTICAL DESIGN

AND10059	Fittings Style Of Pipe Threaded, Flared Tube And Hose

(Copies of specifications and standards required by manufacturers in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

2.1.2 Other publications. The following document(s) form a part of this specification to the extent specified herein. The issues of the documents which are indicated as DoD adopted shall be the issue listed in the current DoDISS and the supplement thereto, if applicable.

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

ASTM D 3951-82	Standard Practice For Commercial Packaging

(Application for copies of ASTM publications should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, Pennsylvania 19103.)

2.1.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence.

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3. REQUIREMENTS

3.1 First article testing. This document provides for preproduction (First Article) testing.

3.2 Components. The tester shall consist essentially of a test fluid reservoir, air powered hydraulic pump, manual control valves, pressure regulator, gages, filters, interconnections and adapters.

3.3 Suitability. The tester shall be suitable for its intended purpose with respect to utility and maintainability, reliability and personnel safety.

3.4 Material. All material used shall be as specified herein. Material not specified shall be suitable for the purpose intended. The material shall be free from all defects and imperfections that might affect the serviceability of the finished product. All metals shall be of a corrosion resistant type or shall be suitably protected to resist corrosion during the normal service life of the unit.

3.5 Design. The tester shall withstand 1.5 times the tester maximum operating pressures. The tester dimensions shall not exceed 24 inches long by 28 inches high by 16 inches wide. The tester dry weight shall not exceed 75 pounds. The tester working fluid shall be MIL-H-5606 hydraulic fluid. Seals and gaskets wetted by hydraulic fluid shall be intended for use with MIL-H-5606 oil. The tester shall be rigged so as to withstand long time service in a general repair shop. Maximum pneumatic operating pressure is defined as 110 psig.

3.6 Performance. The tester shall proof pressure test hose assemblies conforming to MS28741-3 to -32 and hose assemblies conforming to MS28762-4 to -16. The tester shall produce and deliver hydraulic pressure up to 9,000 psig when connected to a source of 90 psig ship air. Tester manual controls shall be such that test hydraulic pressure can be easily increased by the operator at any rate between 15,000 to 35,000 psig per minute. The tester shall function without either oil or air leaks. The tester shall fill hose assemblies on test. Time to fill a 36-inch long MS28741-32 hose assembly shall not exceed one minute. The tester shall air bleed hose assemblies on test. The tester shall drain hose assemblies on test. The tester when operating in any mode or under any load condition shall be no noisier than 80 dBA at five feet from the tester in any direction.

3.7 Details of components.

3.7.1 Hydraulic components.

3.7.1.1 Adapters. Adapters shall be furnished to connect one end of hose assemblies on test to the tester and to plug the other end of the hose assemblies liquid tight. Adapters shall be furnished as required for MS28741-3 to -32 hose assemblies and MS28762-4 to -16 hose assemblies. O-ring preformed packing shall be furnished as required. Each adapter shall be clearly and permanently marked as to its identity.

3.7.1.1.1 Adapter storage. Storage for the adapter shall be furnished integral with the tester. A specific storage location for each adapter shall be furnished. Each adapter storage location shall be clearly and permanently marked as to the identity of the adapter to be stored there.

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3.7.1.2 Fittings. Wherever possible, interconnecting fittings shall be selected from AND10059, and threads, where required, shall conform to MIL-S-7742 and MIL-P-7105.

3.7.1.3 Hydraulic filter. A hydraulic filter shall be installed between the reservoir and the suction side of the pump. The filter element shall be easily removed and replaced. The element shall conform to AN6236-1 except that the element shall pass no particle 10 micron or larger.

3.7.1.4 Hydraulic gages. The tester shall be fitted with two gages to display the hydraulic output pressure of the pump. Each gage shall conform to Class 1, Type 1 of GG-G-76 with adjustable red marking pointer. One gage shall have a scale range of from 0 to 3000 psig. The 0-3000 gage shall be fitted with a gage saver to protect the gage when pressure exceeds 3000 psig. The other gage shall have a scale range of 0-15,000 psig.

3.7.1.5 Hose filling. Means shall be provided to fill test hose assemblies with low pressure hydraulic fluid.

3.7.1.6 Air bleed. Means shall be provided for bleeding all air out of the tester hydraulic test circuit and hose assembly on test. A visual (sight glass or sight tube) means shall be furnished that permits observation of the bleed air bubbles. Hydraulic fluid removed during air bleeding shall return to the reservoir.

3.7.1.7 Hose drain. Means shall be provided for draining hydraulic oil out of hose assemblies and returning the oil to the reservoir upon completion of proof pressure testing of hose assemblies.

3.7.1.8 Reservoir. A reservoir shall be provided. The reservoir shall have a useful capacity of not less than 5 gallons. It is contemplated that most other tester components will be mounted on the reservoir; if this be true, the reservoir shall have adequate strength and stiffness to support the mounted components and to withstand the dynamic and static load incident to the testing of hose assemblies. Interior of the reservoir shall have no pockets or sharp edges and shall be easy to wipe clean. The reservoir shall be provided with a cleanout hole not less than 6 inches in diameter. A filler cap with a 10 micron absolute vent filter shall be provided. The reservoir filler opening, or neck, shall be provided with a removable 30 mesh conical strainer. The reservoir shall be provided with a fluid level sight gage marked to indicate the full level. The hydraulic filter (paragraph 3.7.1.3) shall be located within the reservoir.

3.7.1.9 Hydraulic valves. A fill valve to control filling of hose assemblies on test shall be provided. A bleed valve to control air bleeding of the tester hydraulic test circuit and hose assembly on test shall be provided. A hose drain valve to control return of hydraulic fluid from hose assemblies on test to the reservoir shall be provided. A unit drain valve to control draining of the tester shall be provided. The foregoing valve operations may be combined so that one valve controls more than one specified function.

3.7.1.10 Packings. Wherever possible packings and gaskets which conform to MIL-P-5510 or MIL-P-5516 as applicable and, if feasible, shall be installed in accordance with MIL-P-5514.

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3.7.2 Pneumatic components. Pneumatic components shall properly serve and operate when the tester is connected to a source of 90 to 100 psig shop compressed air.

3.7.2.1 Air filter. A 10 micron absolute air filter shall be provided at the air inlet connection.

3.7.2.2 Air regulator. An adjustable air regulator valve shall be provided in the air inlet line to control the hydraulic outlet pressure between 300 and 10,000 psig. (The 10,000 psig output pressure occurs when the 9,000 psig pressure limiting device is purposely inoperative.)

3.7.2.3 Air lubricator. A lubricating device shall be installed in the inlet line between the regulator and the pump to supply lubrication at the rate of 3 to 4 drops per minute to the pump.

3.7.2.4 Air valve. A shutoff valve shall be provided in the air line to the pump to shut off air flow to the pump.

3.7.2.5 Air gage. The tester shall be provided with an inlet air pressure gage conforming to Class 1, Type I, of GG-G-76 with a scale range of 0-200 psi.

3.7.2.6 Air inlet. The air inlet connection shall be 3/8 inch NPT female.

3.7.3 Pump. An air operated hydraulic pump shall be furnished. The pump, with pressure limiting device inoperative, shall produce not less than 10,000 psig pressure when connected through the paragraph 3.7.2 components to a source of 90 psig shop air. The pump shall be fitted with a nonadjustable device that shall limit pump output to 9,000 psig. The pressure limiting device shall not dump oil to the exterior of the tester.

3.7.4 Legs. The tester shall be furnished with support legs or mounting flanges. Legs shall be provided with holes for securing of the tester to a bench top. Legs shall have adequate strength and stiffness to support the tester and to withstand the dynamic and static loads incident to testing of hose assemblies.

3.8. General. The tester shall comply with the General Specification for the Design of Special Aeronautical Support Equipment, MIL-S-8512, to the extent of the particular paragraphs listed below:

3.1 General.

3.1.1 Conflict in requirements.

3.1.2 Selection of specifications, standards and drawings.

3.1.3 First article.

3.2.1 Selection of materials, parts and processes.

3.2.2 Magnesium.

3.2.3 Dissimilar metals.

3.2.5 Corrosion resistance.

3.2.6 thru 3.2.6.4 Standard parts.

3.2.7 Interchangeability.

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3.3 Design.

3.3.1 System safety engineering (minimum requirement applies).

3.3.1.2 Sharp edges.

3.3.1.5 Safetying.

3.3.7 Maintenance provisions.

3.3.7.1 Accessibility.

3.3.7.3 Disassembly provisions.

3.3.9.1 General design considerations.

3.3.9.2 Rated load.

3.3.9.3 Design load (MIL-A-8421 and MIL-STD-810 do not apply).

3.3.9.5 Allowable stress.

3.4 Construction.

3.4.1 Shock and vibration resistance.

3.4.4 Fastener.

3.4.6.8 Hydraulic systems.

3.4.6.9 Pneumatic systems.

3.7 thru 3.7.2.1 Surface finish.

3.8 thru 3.8.1.4 Nameplate and product marking.

3.8.3.4 Safety marking.

3.8.3.5 Identification marking.

3.8.3.6 Service marking.

3.9 Workmanship.

3.9.1 Certification of welders.

Table II Production processes.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

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4.2 Classification of inspections. The inspection of equipment shall be classified as follows:

a. First article inspection (see 4.4).
b. Quality conformance inspection (see 4.5).

4.3 Test conditions. Unless otherwise specified herein, the following standard conditions shall be used to establish normal performance requirements and for making laboratory tests on the equipment:

Temperature	25° ± 20°C (77° ± 11°F)
Relative humidity	50 ± 30 percent
Atmospheric pressure	725 + 50 mm. Hg. (28.5 + 2.0 in. Hg.) - 75 - 3.0
Fluid	MIL-H-5606
Air pressure	90 psig + 0, -10 psig

4.4 First article testing.

4.4.1 Test sample. Prior to production, one test sample representative of the production item and constructed in accordance with specification requirements, shall be subjected to the first article inspection.

4.4.2 Test report. Upon completion of the first article tests, a test report shall be prepared in accordance with MIL-STD-831. The test report shall be authenticated by a responsible officer of the contractor and shall be countersigned by a government witness. The test report shall contain the following appendixes:

a. Five-view (front, sides, top, rear) drawing of the tester.

b. Four eight by ten inch black and white photographs of the tester (left and right, front and rear three-quarter views).

c. Hydraulic-pneumatic schematic with explanation of operation.

d. Original manufacturer's certification that gages comply with GG-G-76 as specified herein.

e. Original manufacturer's certification that hydraulic fluid reservoir filter will pass no particle 10 micron or larger.

f. Original manufacturer's certification that filler cap air vent filter will pass no particle 10 micron or larger.

4.4.3 First article, or preproduction tests, shall consist of all tests described under test method (see 4.6).

4.5 Quality conformance inspection. Quality conformance, or acceptance, inspection shall consist of the following:

a. Examination of tester (see 4.6.1).

b. Proof pressure test (see 4.6.2).

c. Functional test (see 4.6.3).

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4.5.1 The following paragraphs of MIL-S-8512 apply.

a. 4.5.3 Rejection and retest.

b. 4.5.4 Defects in items already accepted.

4.6 Test methods. Tests shall be performed in the sequence indicated below.

4.6.1 Examination of tester. Examination of equipment shall be conducted to confirm adherence to such requirements as, use of specified components, assemblies, materials finish and marking; adherence to dimensional and weight provisions, workmanship, safety and maintainability provisions. This examination shall consist of visual inspection of equipment, components, and assemblies, simple measurement of dimensions and weight, inspection of production methods, processes, examination of certificates of compliance on purchased materials and components.

4.6.2 Proof pressure test. The tester pneumatic components shall be hydrostatically tested at 165 psig for ten minutes. The tester pump and hydraulic components downstream of the pump shall be hydrostatically tested at 15,000 pounds for ten minutes. The tester hydraulic component, upstream of the pump, including the reservoir, shall be hydrostatically tested at 8 psig for ten minutes. Either leaks or permanent deformation due to the hydrostatic test pressure shall be cause for rejection.

4.6.3 Functional test. Two each 36-inch long hose assemblies conforming to MS28741-3 to-32 and MS28762-4 to -16 shall be filled air bleed, proof pressure tested and drained. The time to perform each of these operations shall be recorded in the test report. This specification imposes no quantitative requirement on any of the foregoing operation except filling of the MS28741-32 hose assembly. Failure to fill the 36-inch long MS28741-32 hose assemblies within one minute shall be cause for rejection. Evidence of malfunction, leaks at tester to hose assembly connections or other undesirable conditions or characteristics shall be cause for rejection.

4.6.4 Control test. Ease of manually controlling the output pressure rise at any rate from 15,000 to 35,000 psig per minute shall be demonstrated.

4.6.5 Noise test. The tester shall be operated in all modes while sound level measurements are made at all points five feet distant from the tester. Any reading greater than 80 dBA shall be cause for rejection.

4.6.6 Shock test. The tester shall be subjected to Procedure V bench handling test of Method 516.1 shock of MIL-STD-810. The requirement that the chassis and front panel assembly be removed from the enclosure as for servicing does not apply. The tester shall be shock tested uncrated, not boxed or otherwise protected. The tester shall be shock tested in a ready-to-use condition except that shop air connection shall not have been made. The tester shall be shock tested with its reservoir filled. At the completion of shock testing, the tester shall be examined for permanent deformation. Permanent deformation of the tester shall be cause for rejection. After inspection of the tester, the tester shall be subjected to the proof pressure test of paragraph 4.6.2 herein. Failure to pass the proof pressure test shall be cause for rejection.

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5. PACKAGING

5.1 Preservation. Preservation shall be level A or Commercial Level as specified (see 6.2).

5.1.1 Level A. Each tester shall be preserved method 1A14 of MIL-P-116.

5.1.2 Commercial. The commercial preservation of each tester shall be in accordance with the requirements of ASTM D 3951-82.

5.2 Packing. Packing shall be level A, B or Commercial as specified (see 6.2).

5.2.1 Level A. The tester packaged as specified in 5.1.1 shall be packed in a wood container conforming to PPP-B-601, overseas type. Closure shall be in accordance with the appendix thereto.

5.2.2 Level B. The tester packaged as specified in 5.1.1 shall be packed in fiberboard containers conforming to PPP-B-636 weather-resistant. Closure shall be in accordance with the appendix therto.

5.2.3 Commercial. The preserved tester shall be packed in accordance with the requirements of ASTM D 3951-82.

5.3 Physical protection. The tester shall be cushioned, anchored, blocked and braced in accordance with MIL-STD-1186. The Freefall Drop Test, Edgewise Drop Test and Crosswise Tests shall be in accordance with Appendix A of MIL-STD-1186. The tests are not required when commercial packing is specified.

5.4 Marking.

5.4.1 Levels A and B. In addition to any special or other identification marking required by the contract (see 6.2), each unit pack, intermediate and exterior container and unitized load shall be marked in accordance with MIL-STD-129.

5.4.2 Commercial. Commercial marking shall be in accordance with the requirements of ASTM D 3951-82.

6. NOTES

6.1 Intended use. The tester is intended for proof pressure testing of hydraulic, fuel and oil hose assemblies to be installed on aircraft and missiles. The tester is also suitable for testing of tube assemblies.

6.2 Ordering data.

6.2.1 Acquisition requirements. Acquisition documents should specify the following:

a. Level of preservation, packaging and packing required (see 5.1.1 and 5.2.1).

b. Contractor/Order special marking requirements (see 5.4).

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions - Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-T- 87971	TESTER, HOSE ASSEMBLIES: HYDRAULIC, AIRCRAFT AND MISSILE

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR
b. ADDRESS *(Street, City, State, ZIP Code)*	☐ USER
	☐ MANUFACTURER
	☐ OTHER *(Specify)*: _____

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

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